



Systematic review and meta-analysis on the association between phthalates exposure and insulin resistance

Bahareh Shoshtari-Yeganeh¹ · Maryam Zarean¹ · Marjan Mansourian² · Roya Riahi³ · Parinaz Poursafa¹ · Hakimeh Teiri¹ · Nasim Rafiei¹ · Bahare Dehdashti¹ · Roya Kelishadi³

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Abstract

This study aims to provide an overview of human studies on the association of exposure to phthalates and insulin resistance. We systematically searched human studies available until 15 January 2018. We conducted a literature search in Scopus, ISI Web of Science, PubMed, Google Scholar, and Cochrane Collaboration. We used the following keywords to identify relevant articles: “phthalate”, “phthalate ester”, “metabolic syndrome”, “insulin resistance”, “glucose intolerance”, and “diabetes”. For analyzing data, we conducted meta-analysis using the Stata software. We appraised each study to examine the sources of heterogeneity, including difference in clinical outcomes and exposure measurements. To determine the robustness and whether some of the factors have the highest impact on the results of the present meta-analysis, several sensitivity analyses were conducted. Sensitivity analysis showed that by removing studies with the highest weight and age groups, no change was observed in heterogeneity. Moreover, with excluding the study conducted in Europe, the results remained unchanged and constant. In addition, the funnel plot and Egger’s tests were executed to access publication bias. Both the funnel plots and Egger’s test did not show any evidence of publication bias ($P=0.31$). In the random effects meta-analysis of all studies ($n=8$), the pooled correlation coefficient between phthalate exposure and HOMA-IR was 0.10 (95% CI; 0.07–0.12, $P<0.001$), with significant heterogeneity ($P<0.001$, $I^2=85.5\%$). Our findings revealed positive association between exposure to phthalate metabolites and increased HOMA-IR; this association remained significant even after adjusting the analysis for multiple confounding variables.

Keywords Phthalates · Endocrine disruptor chemicals · Insulin resistance · Diabetes · Meta-analysis

Introduction

In recent years, one of the important concerns is the widespread use of chemical and industrial products all over the world (Dalsenter et al. 2006). Dialkyl or alkylaryl phthalic acid esters produce compounds that are called phthalates

which are chemically classified into low molecular weight (diethyl phthalate [DEP], dimethyl phthalate [DMP], and dibutyl phthalate [DBP]) and high molecular weight (di(2-ethylhexyl) phthalate [DEHP], diisodecyl phthalate [DIDP], diisononyl phthalate [DINP], and benzyl butyl phthalate [BBP]) (Zarean et al. 2016). Their manufacture began in 1920, and was developed by the production of polyvinyl chloride (PVC) in the 1950s (Kimber and Dearman 2010). These compounds are widely used in cosmetics, toys, pipes, plastic bottles, medical products, and food packaging (Thompson et al. 2009; Zarean et al. 2018). Large amounts (about 90%) of phthalate compounds with high molecular weight (phthalates with more than 6 carbons in their structure, which gives them high durability) are used as plasticizers, while the most components of solvents, inks, pharmaceuticals, cosmetics, and waxes are low molecular weight phthalates (North et al. 2014). For instance, di-2-ethyl hexyl phthalate (DEHP) is one of the high molecular weight phthalates, which is important because of its extensive application in industrial

Responsible editor: Philippe Garrigues

✉ Parinaz Poursafa
parinazpoursafa@gmail.com

¹ Environment Research Center, Research Institute for Primordial Prevention of Non-Communicable Disease, Isfahan University of Medical Sciences, Isfahan, Iran

² Department of Biostatistics and Epidemiology, Faculty of Health, Isfahan University of Medical Sciences, Isfahan, Iran

³ Child Growth and Development Research Center, Research Institute for Primordial Prevention of Non-Communicable Disease, Isfahan University of Medical Sciences, Isfahan, Iran

processes and food packaging (Attina and Trasande 2015). Moreover, phthalates make up 40 to 50% by weight of plastic products that make them durable and flexible (Lind et al. 2012). The main transmission routes of this compound are food, water, dust, and skin contact; however, food is the most important one (Ventrice et al. 2013). In recent decades, some adverse health effects of these chemicals are reported, including organ damages, abortion, cancer, endocrine disruption, and insulin resistance (Reeves et al. 2019). In particular, some endocrine disruptor chemicals (EDCs), including phthalates, are supposed to affect insulin resistance (Hectors et al. 2013).

Insulin resistance is the state of lack of proper response of tissues to insulin (Tsatsoulis et al. 2013). In other words, it occurs when the amount of insulin needed to regulate the transport of plasma glucose to the peripheral tissues increases. The homeostatic model assessment (HOMA-IR) is the usual method used to quantify insulin resistance (Smerieri et al. 2015). Insulin resistance is considered as an important health issue with major roles in the development of type 2 diabetes as well as the metabolic syndrome. In past decades, the number of people involved with these multifactorial disorders has increased very rapidly (Smerieri et al. 2015). On the other hand, diabetes may induce nephropathy, retinopathy, stroke, peripheral vascular disease, and myocardial infarction (Dales et al. 2018). Insulin resistance can lead to cardiovascular disorders by stimulating high blood pressure, glucose intolerance, and dyslipidemia (Amin et al. 2018). Higher body mass index and lower physical activity are major risk factors for insulin resistance; however, chemical such as phthalates may play a role (Dales et al. 2018). There are evidences that phthalates may have an adverse effect on insulin resistance and obesity (Attina and Trasande 2015).

Data regarding the relationship between phthalates exposure and insulin resistance have not been fully elucidated, while some studies showed significant effects and some others did not. To the best of our knowledge, no systematic review or meta-analysis has considered this issue. The purpose of the present study is to conduct a meta-analysis to evaluate the association between phthalates exposure and insulin resistance.

Materials and methods

Search strategy and selection criteria

Search strategy

The systematic search processes were conducted and reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) Statement. We conducted a literature search in Scopus, ISI Web of Science, PubMed, Google Scholar, and Cochrane

Collaboration for articles published before 15 January 2018. We used the following keywords to identify relevant articles: “phthalate”, “phthalate ester”, “metabolic syndrome”, “insulin resistance”, “glucose intolerance”, and “diabetes”. The search was refined to human studies and the English language. Finally, the reference list of the published studies was checked for additional studies.

Eligibility criteria

After the literature search, the identified studies were selected on the basis of the following criteria: (1) studies investigating the associations between exposure to phthalates and effects on insulin resistance; (2) studies with cohort, cross-sectional, or case–control design; and (3) human studies. Those studies that did not report original results, i.e., letters, conference abstracts, comments, or editorials were excluded.

Study selection and data extraction

Two reviewers independently screened all the titles and abstracts of papers according to the predetermined selection criteria and relevant papers were selected, and duplicates were removed. Then, for selecting the included studies, two independent researchers assessed the full texts of potentially relevant studies. An expert investigator resolved the discrepancies. Two reviewers read the reports and extracted the data. The results of this step were compared between the reviewers, and consensus was obtained before the meta-analysis process. The required information that was abstracted from all eligible studies was as follows: data on the first author’s last name, year of publication, location, study design, study population, age, type of phthalate, exposure assessment method, outcomes, risk estimates with their corresponding confidence intervals (CIs), and controlled variables. The same two independent reviewers extracted the most relevant estimators, namely β -coefficients and 95% confidence intervals (CIs).

Quality assessment

The quality of all included studies was evaluated independently and by using a Cochrane checklist. We assessed the study objective, study design, sampling method, sample size, studied target group, data collection tool, and the status of variable evaluation. The analyses status was evaluated by using 12 questions by considering one score for each question (Moosazadeh et al. 2014). At the final step, those studies that obtained at least the acceptable score were selected. Agreement between two reviewers was good (kappa statistics = 0.69 and percent of inter-observed agreement = 85.7%).

Data analysis

We conducted meta-analysis using the Stata software, version 11.2 (STATA Corp, College Station, TX, USA). All *P* values were two-sided with a significance level of less than 0.05. We appraised each study to examine the sources of heterogeneity, including difference in clinical outcomes and exposure measurements. Beta coefficients with corresponding CIs were extracted from the most adjusted model. Standardized beta coefficients from individual studies were used as correlation statistics based on previous research showing that the relationship between a beta coefficient and correlation coefficient is linear, having a correlation of 0.84 if the coefficients reside in the interval ± 0.50 . First, beta coefficients were transformed to the correlation coefficient and *r* to *z* Fisher transformation was applied. We produced forest plots to assess the multivariate adjusted effect size (correlation coefficient) and corresponding 95% CIs visually across studies; we used both fixed and random effects models for analysis. The *Z*-test was used to assess the significance of the pooled correlation coefficient, and *P* value of less than 0.05 was considered as significant.

Assessment of heterogeneity

Heterogeneity between studies was tested using Cochran's *Q* statistic, and quantified by calculating the inconsistency index (I^2). If the tests for heterogeneity were statistically significant in all analysis ($I^2 > 50\%$ indicated substantial heterogeneity), the random effects model was used to consider the between-study variability, by the method of DerSimonian and Liard (Higgins 2011). To evaluate the sources of heterogeneity, we conducted a subgroup analysis with phthalate metabolites and the study region (America/Europe or Asia).

Publication bias

Potential publication bias was tested with funnel plots and both Begg's and Egger's tests. The funnel plot was made by the natural logarithm of the estimate of effect size versus the inverse of standard error (1/SE). *P* value < 0.05 indicated statistically significant publication bias. By using the linear regression method, we explored funnel plot asymmetry that might result from the non-publication of small studies without statistically significant effects.

Sensitivity analysis

To determine the robustness and whether some of the factors have the highest impact on the results of the present meta-analysis, we conducted several sensitivity analyses including: (a) removing the studies with highest weight, (b) removing the studies with highest age groups, and (c) removing the study conducted in Europe.

Results

Study and participant characteristics

After a comprehensive search through electronic databases, we selected 304 articles that matched our study. From the 304 citations reviewed, 8 articles were included in this systematic review and meta-analysis. Generally, the reasons for exclusion of some of the searched articles were animal study, different outcomes for our study, and weak and unrelated data. The description of search process, screening, and final selection is represented in Fig. 1.

All included studies were cross-sectional. The publication year of studies ranged from 2007 to 2018. Studies were conducted in five different countries: one in Canada, one in Sweden, three in the USA, and four in Asian countries (Taiwan and Korea). The participants' age in selected studies ranged from 12 to 79 years. All studies comprised both male and female genders, except that one of them was conducted only in men. The main characteristics of studies included in this meta-analysis are shown in Table 1.

Sensitivity analysis showed that by removing studies with the highest weight and age groups, no change in heterogeneity was observed. Moreover, with excluding the study conducted in Europe, the results of our analyses remained unchanged and constant. In addition, the funnel plot and Egger's tests were executed to access publication bias. Both the funnel plots (figures not shown) and Egger's test suggested no evidence of publication bias (*P* value = 0.31).

In the random effects meta-analysis of all studies ($n = 8$), the pooled correlation coefficient between phthalate exposure and HOMA-IR was 0.10 (95% CI: 0.07–0.12, *P* value < 0.001), with significant heterogeneity (*P* value < 0.001, $I^2 = 85.5\%$). Estimate of between-study variance (Tau-squared) was 0.01 (Fig. 2). Begg's test showed no publication bias (*P* value = 0.31).

Subgroup met-analysis according to phthalate metabolite and region

Results of significant test of pooled correlation coefficient and heterogeneity test according to the phthalate metabolites and region are shown in Table 2. In the random effects subgroup analysis of phthalate metabolites, the subgroup correlation between exposure to phthalate metabolites and HOMA-IR was significant in 10 subgroups (MBP, MBzP, MCP, MECPP, MEHP, MEOHP, MiBP, MnBP, and total DEHP), but in the subgroup study of MEHHP, MEP, and MMP, the correlation between exposure to phthalate metabolites and HOMA-IR was not significant (Table 2). Heterogeneity was not significant in three subgroup studies including MBP (*P*

Fig. 1 Flowchart for selection of studies eligible for the systematic review/meta-analysis

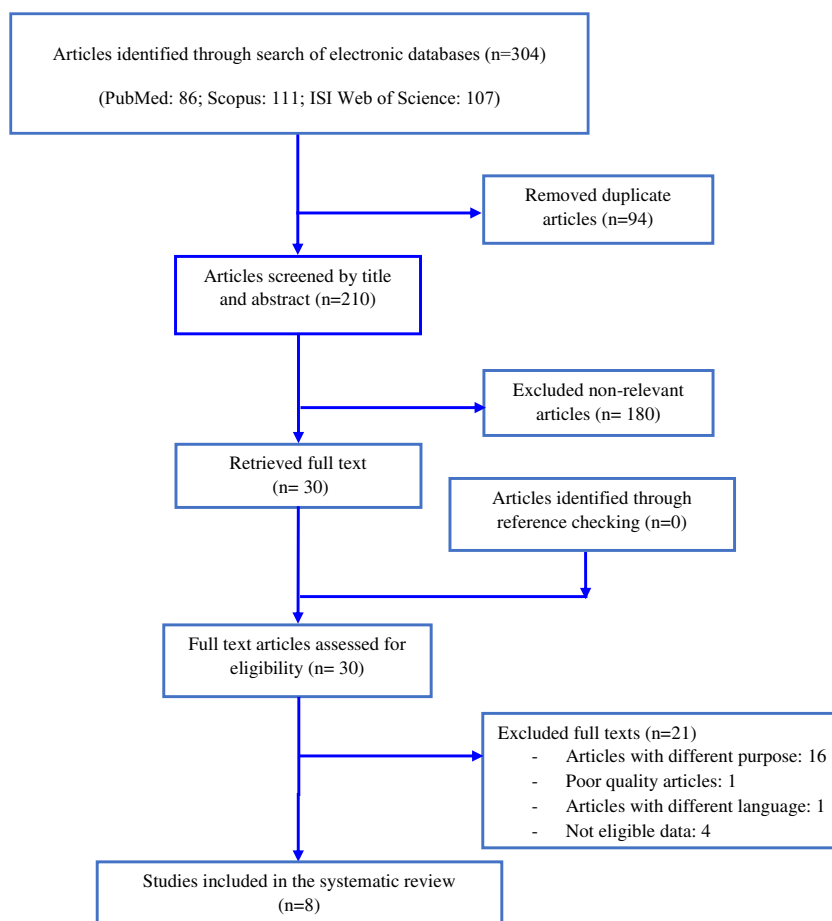


Table 1 Main characteristics of studies included in the systematic review

year	Country	First author	Study design	Sample size	Age group (year)	Confounding factors
2018	Canada	Robert Edgar Dales	Cross-sectional	4437	12–79	Age, sex, race/ethnicity (Caucasian v. other), smoking (current v. other), alcohol (daily over the past 12 months v. other), and physical activity
2017	Taiwan	Szu-Ying Chen	Cross-sectional	786	12–30	Sex, BMI, smoking, alcohol drinking, hypertension, diabetes, total cholesterol, triglycerides, physical activity, hs-CRP, and household income
2015	USA	M Tresa	Cross-sectional	356	12–19	Urinary creatinine, age and caloric intake, sex, PIR, serum cotinine, BMI, race/ethnicity categories, and PA
2013	Korea	Jin Hee Kim	Cross-sectional	560	60 and over	Age, sex, BMI, educational attainment, exercise, and cotinine level, PM ₁₀ on lag day 4, O ₃ on lag day 5, NO ₂ on lag day 7, and outdoor temperature and dew point in the day, total caloric and fat intake
2016	Taiwan	Chien-Yu Lin	Cross-sectional	793	12–30	Age, gender, smoking status
2012	Sweden	P. Monica Lind	Cross-sectional	1016	70	Sex, serum cholesterol and triglycerides, BMI, smoking and exercise habits, and educational levels
2007	USA	Richard W. Stahlhut	Cross-sectional	5094	Over 18	Age, race/ethnicity, total fat and calorie intake, physical activity level, smoking exposure, and urine creatinine, GFR, ALT, and GGT
2013	USA	Leonardo Trasande	Cross-sectional	766	12–19	Urinary creatinine, age and BMI category, gender, PIR, parental education, serum cotinine, and race/ethnicity, caloric intake

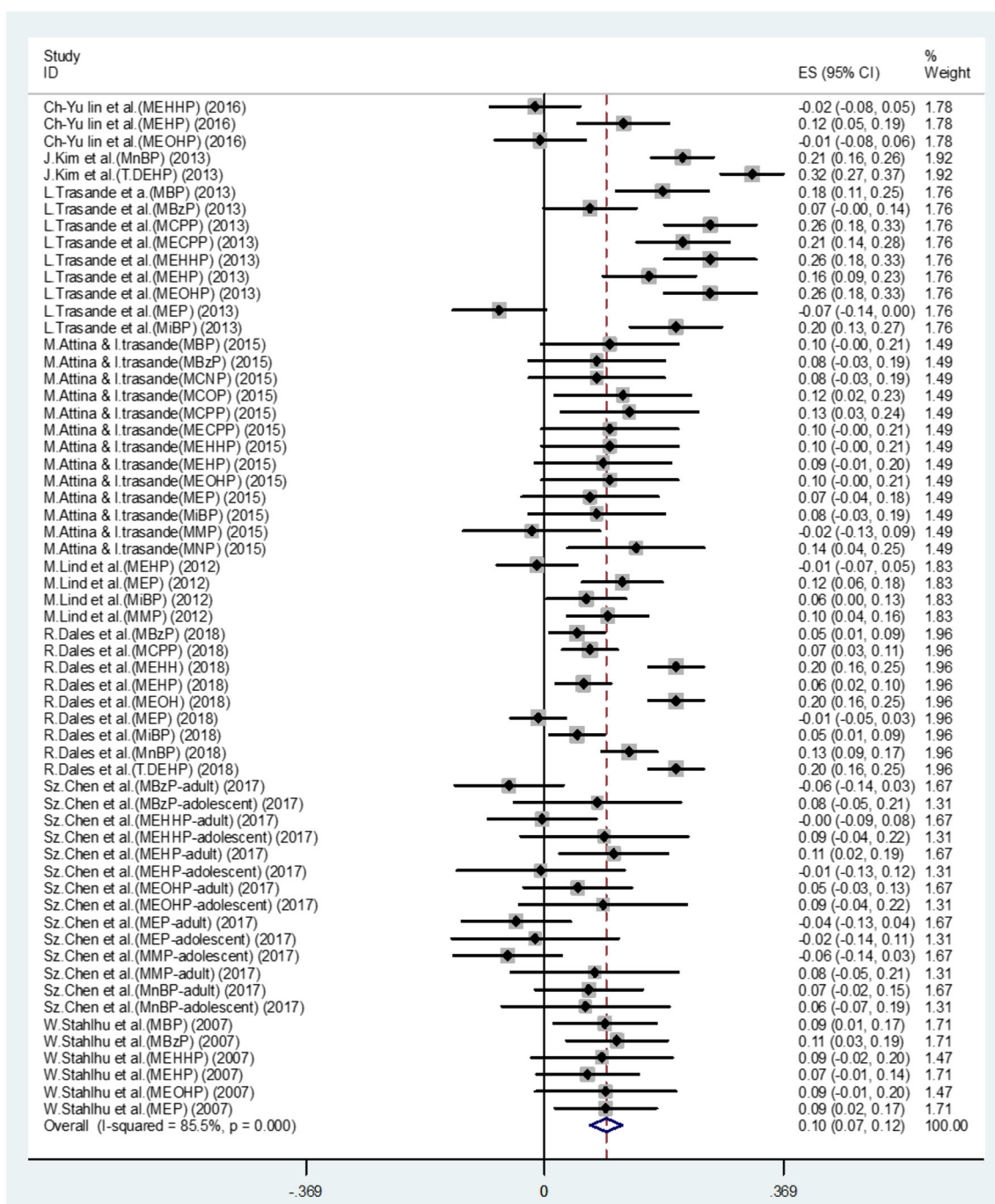


Fig. 2 Association of phthalate exposure with insulin resistance (HOMA-IR). Weights are from random effect analysis

value = 0.21, $I^2 = 36.8\%$), MBzP (P value = 0.10, $I^2 = 45.7\%$), and MECP (P value = 0.08, $I^2 = 67\%$).

In the random effects subgroup analysis of phthalate metabolites, the subgroup correlation between exposure to phthalate metabolites and HOMA-IR was significant in two subgroups of studies in Asia and America and Europe (Table 2). Heterogeneity was significant in two subgroup studies of Asia (P value < 0.001, $I^2 = 89.1\%$), and America and Europe (P value < 0.001, $I^2 = 83.1\%$).

Discussion

The purpose of this study was to determine the evidence of the association of phthalate exposure with insulin resistance and type 2 diabetes in the general population. This systematic review included 11 epidemiological studies; the meta-analysis was conducted on 8 of them that used HOMA-IR as a continuous parameter in a linear regression test. Our findings revealed a positive association between exposure to phthalate

metabolites and increased HOMA-IR; this association remained significant even after adjusting the analysis for multiple confounding variables.

In recent years, growing number of studies have been conducted on the possible role of exposure to EDCs in the development of non-communicable diseases. Given that phthalates are widely used in several products, a considerable part of these studies is related to these compounds. In 2014, for the first time, a systematic review of the epidemiological evidence revealed the obesogenic properties of phthalates (Goodman et al. 2014). It is also documented that in overweight and obese individuals, higher levels of most phthalate metabolites increase the risk of insulin resistance (Dirinck et al. 2015; Lind et al. 2012).

To the best of our knowledge, the current study is the first systematic review and meta-analysis on the association of exposure to phthalates and insulin resistance. Findings of this meta-analyses indicated that phthalate metabolites were weakly associated with increased insulin resistance and therefore the risk of diabetes. Despite differences in the type of metabolites, overall, the articles included in our study showed positive association between exposure to phthalate metabolites and increased HOMA-IR. However, in some studies, different results were reported for various phthalate metabolites. For

instance, in the cross-sectional study of Lin et.al that investigated three types of phthalate metabolites including MEHP, MEHHP, and MEOHP, increased HOMA-IR was shown only for the exposure to MEHP, but the two other metabolites did not have any association (Lin et al. 2016). Two cross-sectional studies linked the levels of these metabolites with increased HOMA-IR (Dales et al. 2018; Trasande et al. 2013), whereas some other studies did not confirm this association (Attina and Trasande 2015; Chen et al. 2017; Stahlhut et al. 2007). MECPP is another type of phthalate metabolite in DEHP group; controversial results exist on its association with insulin resistance. While one study using data from 766 fasting 12- to 19-year-olds in the 2003–2008 revealed this association (Trasande et al. 2013), another study of NHANES 2009–2012 composed of 356 fasting 12–19-year-olds did not find the association (Attina and Trasande 2015). Some studies revealed gender differences in the phthalate metabolite associated with insulin resistance; a study using the US National Health and Nutrition Examination Survey found strong correlation of DEHP metabolites with diabetes in men (Huang et al. 2014) whereas another study using the same data showed this association for MnBP and MiBP with diabetes in women (James-Todd et al. 2012).

Table 2 Test of significant effect size and heterogeneity according to phthalate metabolites

Subgroup	Significance test(s) of ES = 0		Test(s) of heterogeneity		
	Pooled correlation coef. (95% CI)	P value	*I ²	P value	Tau-squared
Phthalate metabolites					
MBP	0.13 (0.07 0.19)	< 0.001*	36.8%	0.21	0.001
MBzP	0.05 (0.01 0.10)	0.02*	45.7%	0.10	0.001
MCPP	0.15 (0.03 0.28)	0.02*	89.6%	< 0.001*	0.011
MECPP	0.16 (0.05 0.27)	0.003*	67.0%	0.08	0.004
MEHHP	0.09 (− 0.01 0.18)	0.08	85.4%	< 0.001*	0.012
MEHP	0.08 (0.03 0.12)	< 0.001*	60.1%	0.01*	0.002
MEOHP	0.10 (0.01 0.19)	0.03*	82.6%	< 0.001*	0.010
MEP	0.02 (− 0.04 0.08)	0.47	75.8%	< 0.001*	0.004
MiBP	0.10 (0.03 0.17)	0.006*	77.8%	0.004*	0.004
MMP	0.02 (− 0.06 0.11)	0.56	70.0%	0.02*	0.005
MnBP	0.13 (0.06 0.20)	< 0.001*	76.5%	0.01*	0.003
T.DEHP	0.26 (0.15 0.38)	< 0.001*	92.2%	< 0.001*	0.006
Overall	0.09 (0.06 0.11)	< 0.001*	85.1%	< 0.001*	0.007
Region					
Asia	0.06 (0.001 0.12)	0.035*	89.1%	< 0.001*	0.014
America and Europe	0.11 (0.08 0.14)	< 0.001*	83.1%	< 0.001*	0.005
Overall	0.10 (0.07 0.12)	< 0.001*	85.5%	< 0.001*	0.007

ES effect size

*Significant at level 0.05

**I²: the variation in ES attributable to heterogeneity

Pooled correlation coefficient is from random effect analysis

Evidence of associations between exposure to low molecular weight group of phthalates and increased HOMA-IR is also inconsistent. A cross-sectional study of 560 participants aged ≥ 60 years in Korea, found a positive and significant association between MnBP and increased HOMA-IR (Kim et al. 2013). Likewise, the study on adult American men found a weak association between MBP and MEP with increased HOMA-IR (Stahlhut et al. 2007). In a study of 766 participants, aged 12 to 19 years in the USA and 4437 individuals, aged 12 to 79 years in Canada, some significant associations were reported between MBP (or MnBP) and MiBP and increased HOMA-IR (Dales et al. 2018; Trasande et al. 2013). However, these associations were not significant in studies conducted among 356 American adolescents and 786 Taiwanese adolescents and young adults (Attina and Trasande 2015; Chen et al. 2017).

Some other studies have investigated the association of high molecular weight group of phthalate metabolites. In a cross-sectional study in Canadian adults (Dales et al. 2018), a very weak association was found between levels of MBzP and MCPP with increased HOMA-IR. Likewise, the findings of a study in the US adult males who participated in the National Health and Nutrition Examination Survey (NHANES, 1999–2002) and another study using data from 766 fasting 12 to 19-year-olds in the 2003–2008 NHANES on MBzP and MCPP respectively showed consistent association (Stahlhut et al. 2007; Trasande et al. 2013). Likewise, in a cross-sectional study of NHANES 2009–2012 that was conducted among 356 adolescents with 12–19 years of age revealed that increased HOMA-IR is related to MCOP, MCPP, and MNP levels (Attina and Trasande 2015).

The underlying mechanism of the association of phthalate exposure with insulin resistance remains to be determined. It is suggested that the activity of phthalate as a ligand for modifying the signals of nuclear receptors involved for metabolism and energy homeostasis regulation might be the underlying mechanism for this effect (Stojanoska et al. 2017). Phthalates target the peroxisome proliferator-activated receptors, which have important functions in regulating fatty acid storage and glucose homeostasis (Minatoya et al. 2017). Therefore, it can be proposed that the binding of phthalates to these receptors might disrupt the regulatory pathways and in turn, can lead to insulin resistance. Oxidative stress might be the other underlying mechanism for the effects of exposure to phthalates on insulin resistance (Kim et al. 2013).

Limitations and strengths of the current review

The findings of the current review should be considered with some limitations. First, selected studies included some differences in study population, types of phthalate metabolites, extracted outcome, and covariates. Despite some differences, overall, the studies showed a positive association between

phthalate exposure and increased HOMA-IR. Second, due to large heterogeneity in using a fixed-effect model, we used the random effects model in the analyses. Some other analyses including subgroup analyses and sensitivity tests were conducted to investigate the heterogeneity source, but none of them could explain the source of heterogeneity. We applied the Begg's test to probe the presence of publication bias across the studies included in this meta-analysis; there was no significant publication bias. Third, this review is limited to studies published in the English language. It should be also acknowledged that although HOMA-IR is usually used as a surrogate marker of insulin resistance, it is not a perfect estimate for insulin sensitivity (Levy-Marchal et al. 2010; Kang et al. 2005). The main strengths of this study are its novelty and the applicable findings that can be useful in the prevention of chronic non-communicable diseases.

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